Data Science Retreat

Project Propsal: AI Music Mentor

Problem: Beginner electronic music producers get stuck finishing tracks because they can't access the kind of subjective, style-specific feedback that experienced producers give; rather than generic technical advice.

Value Opportunity: The electronic music industry ($12.9 billion globally) saw 30% growth in newcomer producers in 2024. A $5/month subscription service targeting beginner techno producers addresses this underserved market at a survey validated price point. Writer's block consistently ranks in the top 3 producer problems across research and online forums.

Solution: Genre-specific AI Mentor **using one experienced producers feedback** to provide track-specific subjective guidance when traditional resources fall short. The AI gives descriptive comparisons between unfinished tracks and reference tracks, plus subjective practical guidance to move songs forward one step with relevant musical language & references.

Project UI: Producer uploads unfinished audio file, reference track (pro finished track close to what they are aiming for), and specific query (e.g., "help extending my arrangement").

Project Output: Text-based analysis with actionable feedback, technical suggestions, and musical reference points tailored to uploaded material and query. Purely text-based to empower human implementation while avoiding copyright issues (eg. avoid using audio generation), topics that were consistently important in survey data.

Technical Approach: LLM Pre-trained on music theory + RAG with audio embeddings and Librosa analysis. Primary focus on arrangement/structure analysis, with eq/rhythm/sound design as stretch goals. Database entries contain {unfinished_song, reference_track, detailed_producer_feedback} with technical analysis and "next-step" improvement suggestions. We leverage RAG to find similarities in audio embeddings and feedback, **delivering specific subjective feedback in the tone and style of one specific producer (eg. myself).** Currently researching: ChatMusician / LLaMA-3-8B / Audio embeddings (CLAP/OpenL3) / MusicBert

Dataset: 50-100 data point triplets from my producer network (songs at multiple development stages, chunked meaningfully to preserve structure), scaling to 200 if needed. Curated text feedback from myself with 12 years experience from music production, running a label and hosting feedback workshops, with plans to incorporate an industry producer post-bootcamp if the prototype goes well.

Success Metrics: 2-3 semi-professional producers blind-test base LLM vs RAG system outputs, rating helpfulness, clarity, and musical accuracy on 1-5 scale.

Key Challenge: Creating meaningful audio embeddings and chunking that capture sequential song structure and genre-specific patterns while maintaining feedback relevance. Meaningfully adapting RAG retrievals to new audio/query without generic output or hallucinations.

Nice to Haves: LLM fine-tuning on genre specific music resources, adjustable feedback tones via system prompts and a Langchain workflow eg. more supportive vs technical feedback.

Paper Reference: Goal to improve on their method **https://arxiv.org/abs/2412.06617**

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20
25